at some point or perhaps allow rotor assemblies to rub blower housing. This condition will usually show up at end of rotors at which bearing has failed.

2. Excessive backlash in blower timing gears usually results in rotor lobes rubbing throughout their entire length.

To correct either of the above conditions the blower must be removed from engine, and either repaired or replaced.

Broken or Cracked Parts

If inspection reveals any defective parts, blower should be repaired before operation.

Noisy or Jammed Blower

If blower develops a harsh metallic sound, clearances have taken up and adjustments or repairs are necessary. If blower does not turn over, do NOT attempt to run engine. Remove blower for overhaul.

Defective Blower Drive

Operate engine at idling speed. Defective coupling will cause loose rattling sound within blower.

NOTE: A loose water pump coupling will sometimes cause a similar noise. Check movement of rotors at lobe tips. Rotors should move ⅜" to ⅝" with a spring-like action. When released, they should spring back at least ¼". If this test indicates necessity, repair coupling as instructed in GEAR TRAIN section of this manual.

Rotor Clearances

Blower clearances are given in figure 24. If rotors bind on feeler in any of the following tests, blower must be removed, disassembled, and repaired. Make preliminary check of rotor lobe clearance as follows:

1. Insert feeler between lobes of rotors as shown in figure 25 for "C" clearance, and as shown in figure 26 for "CC" clearance.

NOTE: When checking "C" clearance, do not insert feeler too far as "CC" clearance will be obtained instead. Also, when measuring clearances greater than .005," laminated feelers made up of .002", .003" or .005" are more practical than one single thick feeler.

2. Turn engine slowly with bar while feeler position is changed to located tight spots. Check clearance at several points along length of each rotor.

3. Now insert feeler between lower rotor and housing to check for "D" clearance.

4. Insert feeler between upper rotor and housing to check for "D" clearance in same manner.

REMOVAL

Removal of the blower assembly together with governor drive, water pump, fuel oil pump, and blower drive shaft cover, will be found advantageous in most cases. For removal of this assembly the following may be used as a guide:

1. Drain water from cooling system as instructed in manual covering respective vehicle.

2. Remove the governor control housing assembly as directed in respective GOVERNOR in **FUEL SYSTEM** section of this manual.

3. Disconnect fuel lines at fuel pump.

4. Remove water pump connections at oil cooler and cylinder block as directed under WATER PUMP in **COOLING SYSTEM** section of this manual.

5. If engine is equipped with the hand control emergency stop, disconnect Bowden wire control at valve shaft lever-located at lower rear corner of blower air inlet housing. If engine is equipped with electrically operated stop solenoid, disconnect wires.

6. Remove air cleaners and blower air inlet housing, including striker plate and blower screen which are mounted between inlet housing and blower.

7. Loosen blower drive tube seal clamp. Remove retaining cap screws at top and bottom of blower housing.

8. Raise front end of blower slightly to clear water pump to oil cooler connection. Then pull blower assembly (including accessories) forward withdrawing blower drive shaft cover and seal on drive gear housing. Remove drive shaft from coupling attached to blower upper rotor timing gear.

DISASSEMBLY

TOOLS REQUIRED

Certain special tools are required for disassembly and assembly of blower. For information relative to vendor of such tools, refer to **SPECIAL TOOLS** section of this manual.

ACCESSORY REMOVAL

After the blower assembly has been removed from engine, the various accessories may be removed as follows, before the blower is disassembled.

1. Remove cap screws attaching fuel pump to blower end cover plate, then remove fuel pump and drive coupling.

2. Remove cap screws attaching blower drive shaft cover to blower cover plate, then remove cover.